

Rigorous coupled-wave analysis of planar-grating diffraction

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A rigorous coupled-wave approach is used to analyze diffraction by general planar gratings bounded by two different media. The grating fringes may have any orientation (slanted or unslanted) with respect to the grating surfaces. The analysis is based on a state-variables representation and results in a unifying, easily computer-implementable matrix formulation of the general planar-grating diffraction problem. Accurate diffraction characteristics are presented for the first time to the authors' knowledge for general slanted gratings. This present rigorous formulation is compared with rigorous modal theory, approximate two-wave modal theory, approximate multiwave coupled-wave theory, and approximate two-wave coupled-wave theory. Typical errors in the diffraction characteristics introduced by these various approximate theories are evaluated for transmission, slanted, and reflection gratings. Inclusion of higher-order waves in a theory is important for obtaining accurate predictions when forward-diffracted orders are dominant (transmission-grating behavior). Conversely, when backward-diffracted orders dominate (reflection-grating behavior), second derivatives of the field amplitudes and boundary diffraction need to be included to produce accurate results.

1. INTRODUCTION

The diffraction of electromagnetic waves by planar gratings has been extensively studied in recent years. These periodic structures have applications in several diverse areas, such as acousto-optics, integrated optics, holography, and spectroscopy. Several different techniques have been used to analyze the diffraction of electromagnetic waves by spatially modulated media. The most common of these methods are the coupled-wave approach¹⁻⁷ and the modal approach.⁸⁻¹⁷ Both of these approaches are capable of being formulated in a rigorous manner without approximations. As such they have been shown to be exact and equivalent.¹⁸ However, until the present work, accurate calculations have been rather difficult and time consuming. This paper presents a straightforward rigorous coupled-wave analysis that is easily implemented on a computer. The method of solution is based on a state-space representation of the governing set of differential equations.

The coupled-wave approach has long been known to offer superior physical insight into wave-diffraction phenomena, and it frequently yields simple analytical results in limiting cases. In this approach several assumptions typically are made in order to obtain solutions. These approximations are (1) neglecting boundary diffraction,^{1-5,7} (2) neglecting the second derivatives of the field amplitudes,^{1-5,7} and (3) retaining only one diffracted wave^{1,2,4,6} (in addition to the transmitted wave). The first two of these approximations are interconnected and cannot be separated for the case of a planar grating. The full modal approach is a rigorous exact analysis, but solutions have been published only for the unslanted pure transmission case in which the grating vector is parallel to the grating surface. Several authors have attempted to analyze general slanted gratings. Kogelnik⁴ and Magnusson and Gaylord⁷ applied the coupled-wave approach, with various assumptions discussed above, to this problem.

Clearly these analyses are not exact, and in some cases they give obviously incorrect results because of the use of these approximations. Bergstein and Kermisch¹² applied the modal approach to slanted gratings, but they assumed a two-wave regime to obtain solutions. Chu and Kong¹⁷ have also presented a generalized modal approach formulation for the general slanted-gratings case. However, they did not present any calculated results for general slanted gratings because the formulation for slanted gratings in the modal approach results in a complicated transcendental relationship that may be difficult to solve in general. This paper presents, for the first time to the authors' knowledge, accurate diffraction results for slanted gratings.

By using the straightforward rigorous method of analysis presented in this paper, it is a simple matter to analyze the precise effect of using the various approximations mentioned above. The effect of neglecting boundary diffraction and second derivatives of field amplitudes is studied by comparing the present theory with multiwave coupled-wave theory.⁷ The effect of neglecting higher-order waves (retaining only one diffracted wave) is analyzed by comparing the present theory and the two-wave modal theory.¹² The combined effect of these approximations is studied by comparing the present theory with the two-wave coupled-wave theory.⁴ These three errors are calculated and presented for the first time for several typical transmission, reflection, and general planar gratings.

The rigorous coupled-wave approach presented here analyzes the diffraction of an electromagnetic plane wave incident obliquely at a planar grating bounded by two different media. In general, these gratings simultaneously exhibit both transmission and reflection behavior, as indicated by the forward-diffracted and backward-diffracted waves shown in Fig. 1. This analysis is applicable (1) to holographic gratings in air or other media ($\epsilon_1 = \epsilon_3 \neq \epsilon_2$), (2) to acousto-optic gratings within a medium ($\epsilon_1 = \epsilon_3 = \epsilon_2$), and (3) to grating couplers such as are used in integrated optics ($\epsilon_1 \neq \epsilon_2 \neq \epsilon_3 \neq \epsilon_1$).

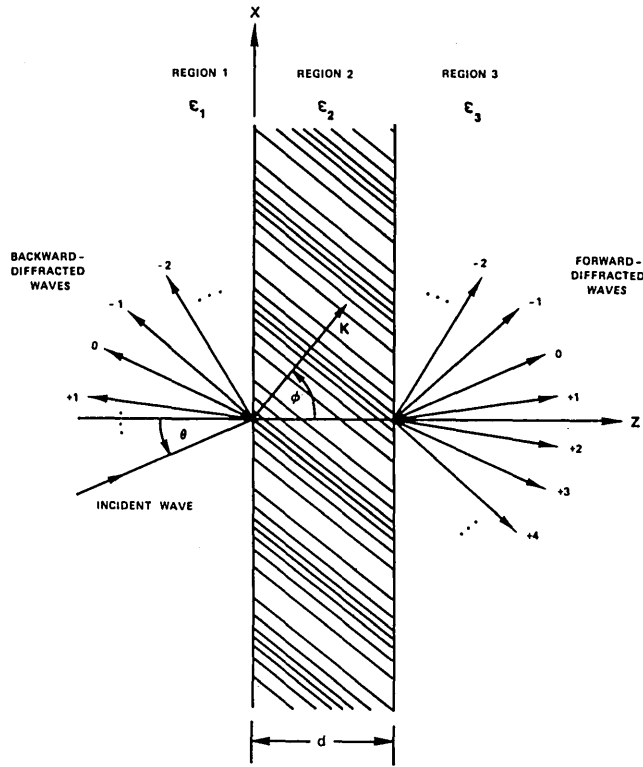


Fig. 1. Geometry for planar-grating diffraction.

2. THEORY

As in most of the work previously cited, the problem under consideration is the diffraction of an obliquely incident plane wave on a lossless (pure phase) sinusoidal grating with the incident wave polarized perpendicular to the plane of incidence (*H* mode). Therefore the electric field will have only one component (in the *y* direction of Fig. 1). The relative permittivity in the modulated region ($0 < z < d$) is

$$\epsilon(x, z) = \epsilon_2 + \Delta\epsilon \cos[K(x \sin \phi + z \cos \phi)], \quad (1)$$

where ϵ_2 is the average dielectric constant, $\Delta\epsilon$ is the amplitude of the sinusoidal relative permittivity, ϕ is the grating slant angle, and $K = 2\pi/\Lambda$, where Λ is the grating period. The dielectric constant in the unmodulated regions ($z < 0$ and $z > d$) is ϵ_1 and ϵ_3 , respectively. It is assumed that each of the three regions has the permeability of free space.

The general approach to the planar-grating problem involves finding a solution of the wave equation in each of the three regions and then matching the tangential electric and magnetic fields at the two interfaces ($z = 0$ and $z = d$) to determine the unknown constants (resulting from solving the differential-wave equation). In region 1, backward-diffracted waves exist. In general, these waves are produced both by diffraction from within the grating volume (bulk diffraction) and by diffraction from the periodic boundary at $z = 0$ (boundary diffraction). These diffraction processes produce a spectrum of plane waves traveling back into region 1 ($z < 0$). The normalized wave amplitudes in region 1 may be expressed as⁸⁻¹⁰

$$E_1 = \exp[-j(\beta_0 x + \xi_{10} z)] + \sum_i R_i \exp[-j(\beta_i x - \xi_{1i} z)], \quad (2)$$

where $\beta_i = k_1 \sin \theta - iK \sin \phi$ for any integer i (the wave

index); $\xi_{li}^2 = k_l^2 - \beta_i^2$ for $l = 1, 3$ (the region index); $k_l = 2\pi\epsilon_l^{1/2}/\lambda$ for $l = 1, 2, 3$; λ is the free-space wavelength; $j = (-1)^{1/2}$; θ is the angle of incidence in region 1; and R_i is the normalized amplitude of the i th reflected wave and is to be determined from the matching of the electric and magnetic fields. In region 3 ($z > d$) the spectrum of transmitted plane waves may be expressed as^{16,17}

$$E_3 = \sum_i T_i \exp[-j[\beta_i x + \xi_{3i}(z - d)]], \quad (3)$$

where T_i is the normalized amplitude of the i th transmitted wave to be determined from the field matching. In region 2, the modulated region ($0 < z < d$), the electric field may be expressed as

$$E_2 = \sum_i \hat{S}_i(z) \exp[-j(\beta_i x + \xi_{2i} z)], \quad (4)$$

where $\xi_{2i} = k_2 \cos \theta' - iK \cos \phi$, θ' is the angle of refraction inside the modulated region, and $\hat{S}_i(z)$ is the normalized amplitude of the i th wave field at any point within the modulated region. For a given value of i , the wave field inside the grating is not a simple plane wave. It may be expressed as a superposition of an infinite number of plane waves (inherent in the coupled-wave formulation). This superposition includes forward-traveling waves (components in $+z$ direction) and corresponding backward-traveling waves (components in $-z$ direction). These amplitudes are to be determined from solving the modulated-region wave equation

$$\nabla^2 E_2 + (2\pi/\lambda)^2 \epsilon(x, z) E_2 = 0. \quad (5)$$

Note that the three electric fields E_1 , E_2 , and E_3 in the three regions are phase matched along the two interfaces. It is important to point out that the above analysis is valid for all slant angles ϕ except when the slant angle is identically zero (pure reflection grating). In this case, the modulation is no longer periodic since it has only a finite number of cycles. Equations (2)–(4) are thus not valid because they are derived from the Floquet theorem, which is correct only for infinite periodic structures. However, the above analysis can be used to analyze pure reflection gratings ($\phi = 0$) by considering $\phi = \delta$ and allowing δ to become arbitrarily small. As ϕ approaches (but does not reach) zero, the slab retains a periodicity in the modulation along its boundaries. All the R_i and T_i amplitudes for the reflected and transmitted waves may then be calculated. As ϕ approaches zero, all the reflected wave vectors in region 1 approach a common direction, and the resulting reflected intensity of the composite wave is $\sum_i |R_i|^2$. Likewise, all the transmitted wave vectors in region 3 approach a single direction, and the resulting transmitted intensity of this composite wave is $\sum_i |T_i|^2$. For a more detailed discussion of planar reflection gratings, see Ref. 19.

3. METHOD OF SOLUTION

To obtain the diffracted amplitudes $\hat{S}_i(z)$, Eqs. (1) and (4) are substituted into Eq. (5), resulting in the infinite set of coupled-wave equations

$$(\Delta\epsilon/8\epsilon_2) \frac{d^2 \hat{S}_i(u)}{du^2} = (\cos \theta' - i\mu \cos \phi) \times \frac{d\hat{S}_i(u)}{du} - \rho i(i - B) \hat{S}_i(u) + \hat{S}_{i+1}(u) + \hat{S}_{i-1}(u), \quad (6)$$

with $S_i(u) = \hat{S}_i(z)$, $u = j\pi\Delta\epsilon z/2\lambda(\epsilon_2)^{1/2} = j\kappa z$, $\mu = \lambda/\Lambda(\epsilon_2)^{1/2}$, $\rho = 2\lambda^2/\Lambda^2\Delta\epsilon = 2\mu^2\epsilon_2/\Delta\epsilon$, and $B = 2\Lambda(\epsilon_2)^{1/2} \cos(\phi - \theta')/\lambda = 2 \cos(\phi - \theta')/\mu$. This system of coupled-wave equations has been derived *without* the common assumptions and approximations associated with previous coupled-wave analyses,¹⁻⁷ such as neglecting the second derivatives of the amplitudes. Therefore the present analysis is as rigorous and as exact as the modal approach. There are five fundamental parameters [θ' , ϕ , $\Delta\epsilon/\epsilon_2$, d/Λ , $\lambda/\Lambda(\epsilon_2)^{1/2}$] contained within Eq. (6). The composite parameters used in Eq. (6) may be expressed in terms of these parameters. Other equivalent sets of parameters obviously may also be chosen. The regime parameter ρ determines the boundary between the Bragg regime and the intermediate diffraction regime.²⁰ The parameter B represents the Bragg condition, i.e. for incidence at the p th Bragg condition $B = p$ (note that B is not an integer for off-Bragg incidence). The parameter κ is the widely used coupling coefficient. As is shown in the Appendix, if S and S' are defined as state variables, Eq. (6) may be written in matrix form as

$$\begin{bmatrix} S' \\ S'' \end{bmatrix} = \begin{bmatrix} b_{rs} \end{bmatrix} \begin{bmatrix} S \\ S' \end{bmatrix}, \quad (7)$$

where S , S' , and S'' indicate the column vectors of S_i , dS_i/du , and d^2S_i/du^2 , respectively. The quantity $[b]$ is the coefficient matrix specified from Eq. (6). Equation (7) corresponds to an unforced state equation in the state-space description of linear systems. The system of differential equations given by Eq. (7) has a relatively simple and straightforward solution, obtainable in terms of the eigenvalues and the eigenvectors of the coefficient matrix $[b]$. It is

$$S_i(u) = \sum_m C_m w_{im} \exp(q_m u), \quad (8)$$

where q_m is the m th eigenvalue and w_{im} is the m th element of the row in the matrix $[w]$ composed of the eigenvectors, corresponding to the i th wave (not the i th row of the matrix $[w]$). The coefficients C_m are unknown constants to be determined together with R_i and T_i by matching the tangential electric and magnetic fields at the two boundaries ($z = 0$ and $z = d$). The four quantities to be matched and the resulting boundary conditions are tangential E at $z = 0$:

$$R_i + \delta_{i0} = \sum_m C_m w_{im}, \quad (9)$$

tangential H at $z = 0$:

$$\xi_{1i}(R_i - \delta_{i0}) = \sum_m C_m w_{im}(q_m \kappa - \xi_{2i}), \quad (10)$$

tangential E at $z = d$:

$$T_i = \sum_m C_m w_{im} \exp[j(q_m \kappa - \xi_{2i})d], \quad (11)$$

tangential H at $z = d$:

$$-\xi_{3i}T_i = \sum_m C_m w_{im}(q_m \kappa - \xi_{2i}) \exp[j(q_m \kappa - \xi_{2i})d], \quad (12)$$

where δ_{i0} is the Kronecker delta function. Eliminating T_i and R_i from these equations gives

$$2\delta_{i0}\xi_{1i} = \sum_m C_m w_{im}(\xi_{2i} - q_m \kappa + \xi_{1i}), \quad (13)$$

$$0 = \sum_m C_m w_{im}(\xi_{2i} - q_m \kappa - \xi_{3i}) \exp(jq_m \kappa). \quad (14)$$

Note that

$$\xi_{2i} = (2\rho\kappa/\mu^2)(\cos\theta' - i\mu \cos\phi), \quad (15)$$

$$\xi_{li} = (2\rho\kappa/\mu^2)[(\epsilon_1/\epsilon_2) - (\sin\theta' - i\mu \sin\phi)^2]^{1/2} \quad l = 1, 3. \quad (16)$$

In regions 1 and 3, ξ_{li} is either positive real (propagating wave) or negative imaginary (evanescent wave). The system of linear equations given by Eqs. (13) and (14) can be solved for C_m , and then R_i and T_i can be calculated from Eqs. (9) and (11). Note that the number of equations available is exactly equal to the number of unknowns. For example, if n waves are retained in the analysis, then there will be n unknown values each of R_i and of T_i and $2n$ unknown values of C_m . This is because the coefficient matrix $[b]$ in Eq. (7) is a $2n \times 2n$ matrix and therefore has $2n$ eigenvalues, and thus there are $2n$ unknown values of C_m . Alternatively, this may be viewed as being due to the n coupled-wave equations, each being a second-order differential equation, and thus there are $2n$ roots or eigenvalues and $2n$ unknown constants C_m to be determined from the boundary conditions. Therefore the total number of unknowns is $4n$, and Eqs. (9)–(12) provide $4n$ linear equations in these unknowns.

To summarize, the algorithm used to solve this problem proceeds as follows: First the coefficient matrix $[b]$ is constructed, and then eigenvalues and eigenvectors are calculated (typically by using a computer library program). The system of linear equations, Eqs. (13) and (14), is then constructed and solved for C_m (using a technique such as gauss elimination). Equations (9) and (11) are then used to calculate the diffracted amplitudes R_i and T_i . Power conservation requires that the sum of the efficiencies for all of the propagating waves be unity. That is,

$$\sum_i (DE_{1i} + DE_{3i}) = 1, \quad (17)$$

where DE_{1i} and DE_{3i} are the diffraction efficiencies in regions 1 and 3, respectively. These diffraction efficiencies are given by

$$DE_{1i} = \text{Re}(\xi_{1i}/\xi_{10})R_i R_i^* \quad (18)$$

and

$$DE_{3i} = \text{Re}(\xi_{3i}/\xi_{10})T_i T_i^*. \quad (19)$$

The real part of the ratio of the propagation constants occurs when the time-average power-flow density is obtained by taking the real part of the complex Poynting vector. The quantity $\text{Re}(\xi_{li}/\xi_{10})$ is just the usual ratio of the cosine of the diffraction angle for the i th wave to the cosine of the angle of the zeroth-order wave for the l th medium. The results of sample calculations for the diffraction efficiencies are shown in Figs. 2–6 as a function of the grating strength parameter γ ($= \kappa d/\cos\theta'$) and as a function of d/Λ for a pure transmission grating ($\phi = \pi/2$), for general slanted gratings ($\phi = \pi/3, \pi/6$), and for a pure reflection grating ($\phi = 0$).

4. COMPARISON WITH PREVIOUS ANALYSES

A. Rigorous Modal Approach

Basically, the rigorous coupled-wave approach presented in this paper and the rigorous modal approach analyze the pla-

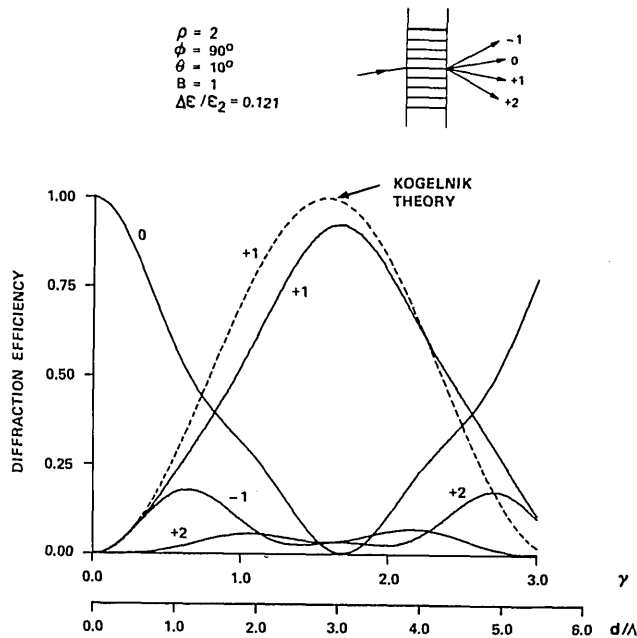


Fig. 2. The diffraction efficiencies of the transmitted waves for pure transmission grating ($\phi = 90^\circ$), with $\rho = 2$. The diffraction efficiencies of all reflected and transmitted waves not shown in the figure are less than 0.01. Here and in Figs. 3-6, $\epsilon_1 = \epsilon_2 = \epsilon_3$.

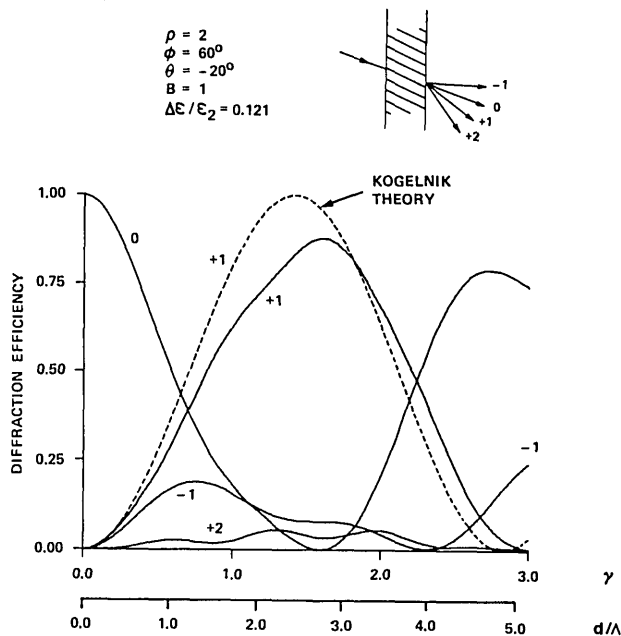


Fig. 3. The diffraction efficiencies of the transmitted waves for a $\phi = 60^\circ$ slanted grating with $\rho = 2$. The diffraction efficiencies of all reflected and transmitted waves not shown in the figure are less than 0.01.

nar-grating diffraction problem by solving the wave equation in the three regions (Fig. 1) and then matching the tangential electric and the magnetic fields at the two boundaries to determine all the unknowns. The main difference between the two approaches is in the technique used to find solutions of the wave equation in the modulated region. In the present coupled-wave approach, the resulting system of coupled-wave equations is formulated into a simple matrix form for which

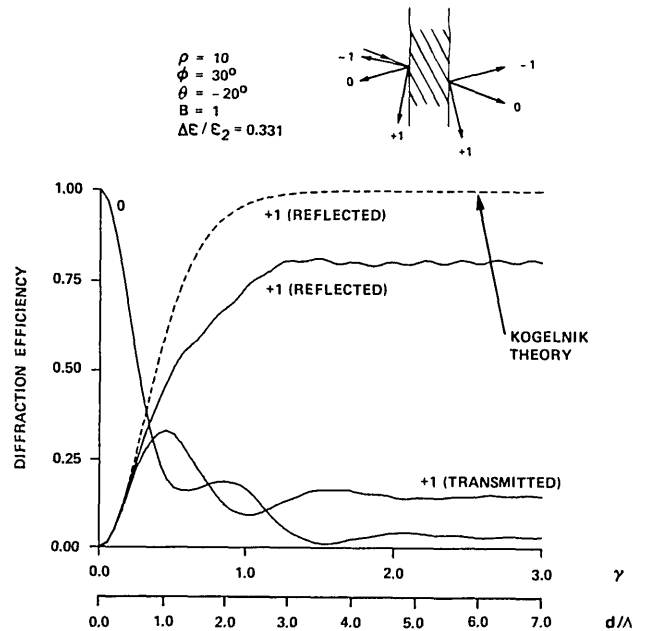


Fig. 4. The diffraction efficiencies for a $\phi = 30^\circ$ grating with $\rho = 10$. The diffraction efficiencies of all reflected and transmitted waves not shown in the figure are less than 0.01.

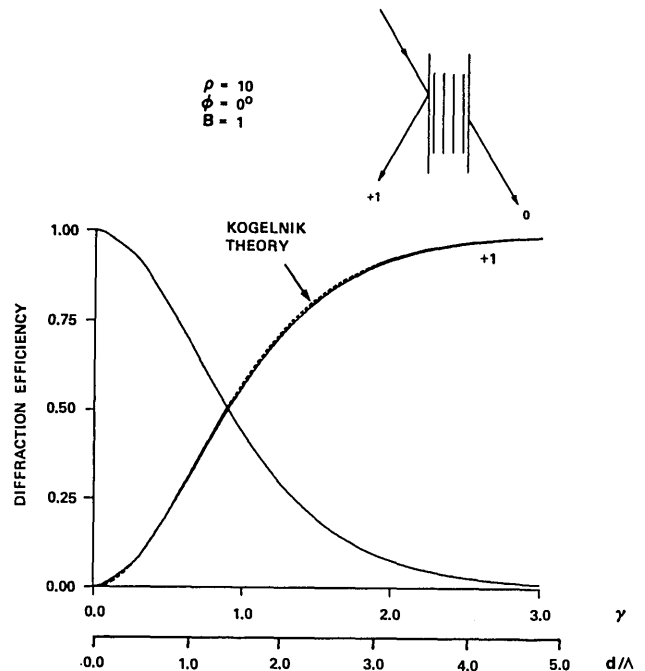


Fig. 5. The diffraction efficiencies of the transmitted and reflected waves at first-Bragg incidence ($B = 1$) for a pure reflection grating ($\phi = 0^\circ$) with $\rho = 10$. These two waves are the only waves that propagate in the unmodulated regions.

the solution is readily obtained by calculating the eigenvalues and the eigenvectors of the coefficient matrix constructed from the coupled-wave equations. The eigenvalue problem, although not simple, has been extensively studied, and numerous efficient and straightforward computer programs to calculate the eigenvalues and the eigenvectors are available in typical computer-program libraries. The modal approach, on the other hand, requires that a transcendental relationship

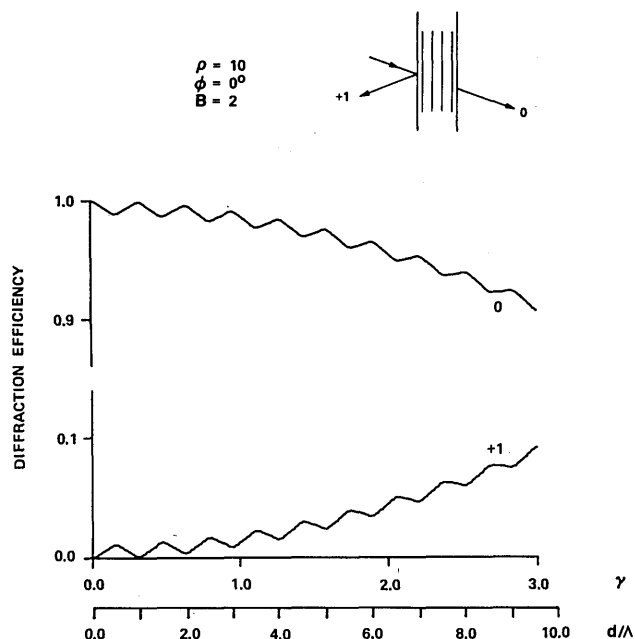


Fig. 6. The diffraction efficiencies of the transmitted and reflected waves at second-Bragg incidence ($B = 2$) for a pure reflection grating ($\phi = 0^\circ$) with $\rho = 10$. These two waves are the only waves that propagate in the unmodulated regions.

in the form of a continued fraction expansion be solved to find the wavenumbers and their corresponding coefficients that are needed to solve the wave equation in the modulated region. This primary difficulty when applying the modal approach to the analysis of slanted gratings is due to the fact that there is no systematic technique yet available to solve this transcendental continued fraction relation for the general slanted-grating case.⁸ For the modal approach the transcendental relationship can also be formulated as a matrix. This matrix is always $n \times n$ (as opposed to $2n \times 2n$ for the present coupled-wave analysis). Only for the unslanted, physically symmetric case is the resulting matrix for the modal approach in standard eigenvalue form. The corresponding wave numbers needed in the solution are the positive and negative square roots of the eigenvalues. The vector of coefficients, which is the same for both the positive and negative wave numbers, is the corresponding eigenvector of the problem. For slanted gratings the resulting modal-approach $n \times n$ matrix is not in the form of a standard eigenvalue problem. Therefore the wave numbers occurring in the matrix cannot be systematically determined.

B. Approximate Modal and Coupled-Wave Analyses

With the exception of the rigorous modal theory, previous analyses have typically been approximate. The present coupled-wave analysis is exact and rigorous, and it reduces to each of the previous analyses with the appropriate approximations and simplifications. For example, it reduces to Kong's coupled-wave analysis⁶ and to Bergstein and Kermisch's analysis,¹² if a two-wave regime is assumed (that is, retaining only the first- and zero-order waves and neglecting all higher-order waves). Kogelnik's⁴ analysis is obtained by assuming a two-wave regime, neglecting boundary diffraction, and neglecting second derivatives of the field amplitudes in Eq. (6). This last approximation, which is common in almost

all previous coupled-wave analyses,^{1-5,7} implies that the parameter $\Delta\epsilon/8\epsilon_2$ was assumed to be very small in these analyses. Magnusson and Gaylord's⁷ analysis is obtained by neglecting boundary diffraction and the second derivatives of the field amplitudes. It is important to note that in previous coupled-wave analyses,^{1-5,7} the amplitudes and intensities of the diffracted waves were calculated inside the modulated region. These analyses are based on solving the half-space grating problem. They do not (and cannot, because of the various approximations) solve the problem of general planar slab grating bounded by two different media.

C. Evaluation of Approximate Approaches

The rigorous coupled-wave theory presented in this paper is easily implemented and permits calculations to be performed rapidly. As such, this analysis is a powerful tool for analyzing the precise effect of using the various assumptions employed in the above approximate methods. It is difficult to quantify the error in general terms. However, the *trends* shown here have been consistently observed in the results of numerous calculations. Figures 7-10 show, for some typical cases, the absolute error in the primary +1 order that is due to using (a) the two-wave coupled-wave theory,⁴ (b) the multiwave coupled-wave theory,⁷ and (c) the two-wave modal theory.¹² In general, these grating structures do not behave simply as transmission gratings or reflection gratings. This is most obvious in slanted gratings (such as that shown in Fig. 4). A combination of transmission and reflection behaviors is typically present, even though one or the other dominates in each case, as determined by the location of the primary +1 diffracted order. For the slanted grating of Fig. 4, the +1 reflected order is the primary +1 diffracted order, and reflection behavior dominates in this case. Note that the +1 transmitted wave (which is phase matched to the +1 reflected wave) would have been assumed to be zero in the two-wave and multiwave (approximate) coupled-wave theories. This nonzero +1 transmitted diffracted wave is attributable entirely to the inclusion of second derivatives in this rigorous theory.

The gratings represented in Figs. 7-10 include the parameter values used in Figs. 2-5. The effect of neglecting boundary diffraction and second derivatives of field amplitudes is determined by the absolute error that results from using multiwave coupled-wave theory. As is shown in Figs. 7 and 8, this error is relatively small when forward-diffracted waves are dominant (transmission-grating behavior). However, when backward-diffracted waves are dominant (reflection-grating behavior), this error becomes relatively large, as is shown in Fig. 9. The effect of neglecting higher-order waves and thus retaining only one diffracted wave is determined by the absolute error that results in using two-wave modal theory. In a complementary way to the preceding results, Figs. 7 and 8 show that this error is large when forward-diffracted waves are dominant, and Fig. 9 shows that it is small when backward-diffracted waves are dominant. The results of many calculations support these conclusions. For example, calculations for two slanted gratings with exactly the same modulation, but one with transmission behavior dominant and the other with reflection behavior dominant, show that higher-order waves are more important in the transmission case and second derivatives and boundary diffraction are more important in the reflection case. Further, these basic conclu-

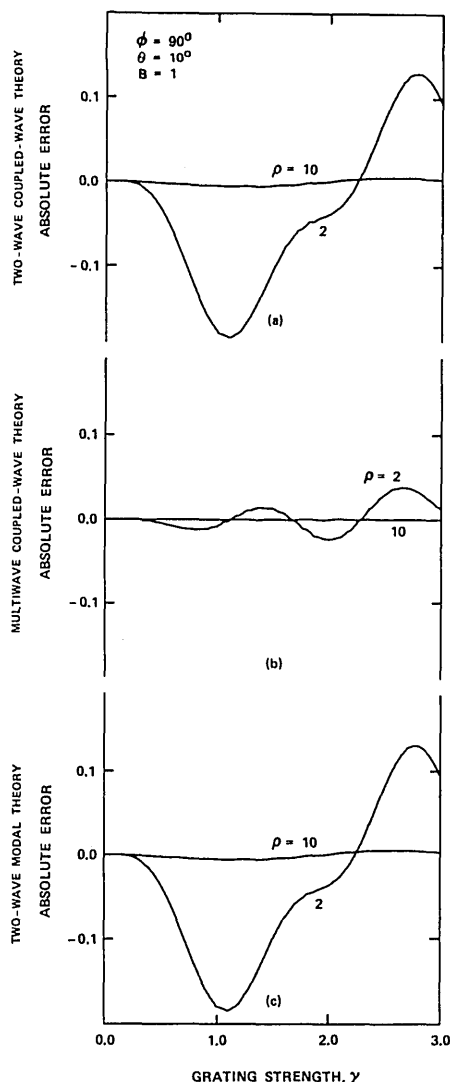


Fig. 7. Absolute error in the first-order diffraction efficiency as predicted by (a) two-wave coupled-wave theory, (b) multiwave coupled-wave theory, and (c) two-wave modal theory for the grating conditions shown in Fig. 2.

sions are supported by calculations for slanted gratings of equal modulation over a wide range of grating modulations. For example, at very small modulations, even though the errors that are due to the neglect of higher-order waves are small and the errors that are due to the neglect of second derivatives and boundary diffraction are small, the importance of these assumptions relative to each other is exactly the same as for large modulations. For equal grating modulations, higher-order waves are more important when transmission behavior dominates, and second derivatives and boundary diffraction are more important when reflection behavior dominates. This relative importance is unchanged regardless of whether the calculated errors are large or small. The combined effect of neglecting both higher-order waves and second derivatives and boundary diffraction is determined by the absolute error that results from using two-wave coupled-wave theory. This error is also shown in Figs. 7–9. From the two-wave coupled-wave theory error, it is consistently found that the total error that is due to neglecting both higher-order waves and second derivatives and boundary diffraction is approximately

equal to the larger of the two constituent errors. The errors shown in Figs. 7–9 are all for the primary +1 order. However, numerous additional calculations for the other waves show a similar dependence on the approximations. For a large variety of cases analyzed, it was found in every case that if higher-order waves or boundary diffraction and second derivatives are important in accurately calculating the primary +1 order, then they are similarly important in accurately calculating the other orders.

For the case of a pure reflection grating ($\phi = 0$), the situation is somewhat more complicated. As ϕ approaches zero, all the forward-diffracted and all the backward-diffracted waves outside the modulated region converge to a single forward and a single backward direction, respectively. The directions of the various orders inside the modulated medium remain distinct. The error associated with the total backward-diffracted wave is shown in Fig. 10. In this case, the errors that are due to neglect of second derivatives and boundary diffraction are similar to those that are due to neglect of higher-order waves.

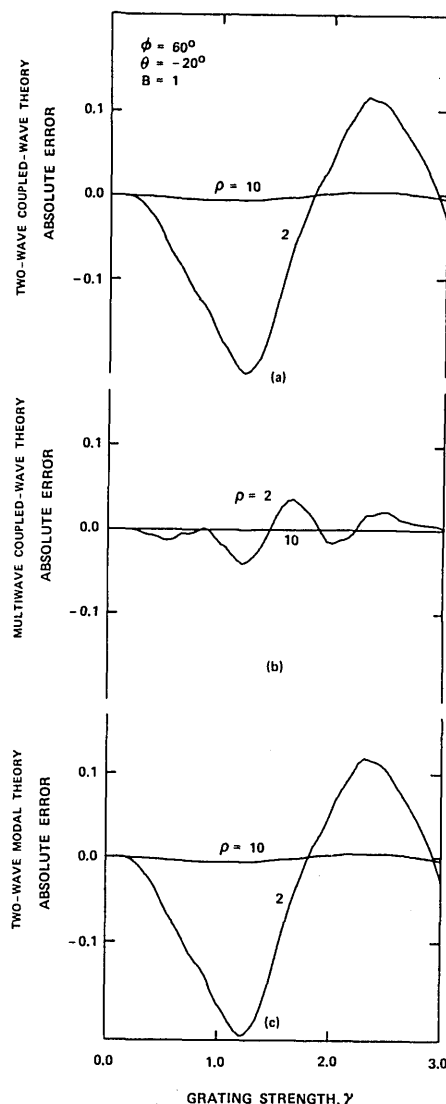


Fig. 8. Absolute error in the first-order diffraction efficiency as predicted by (a) two-wave coupled-wave theory, (b) multiwave coupled-wave theory, and (c) two-wave modal theory for the grating conditions shown in Fig. 3.

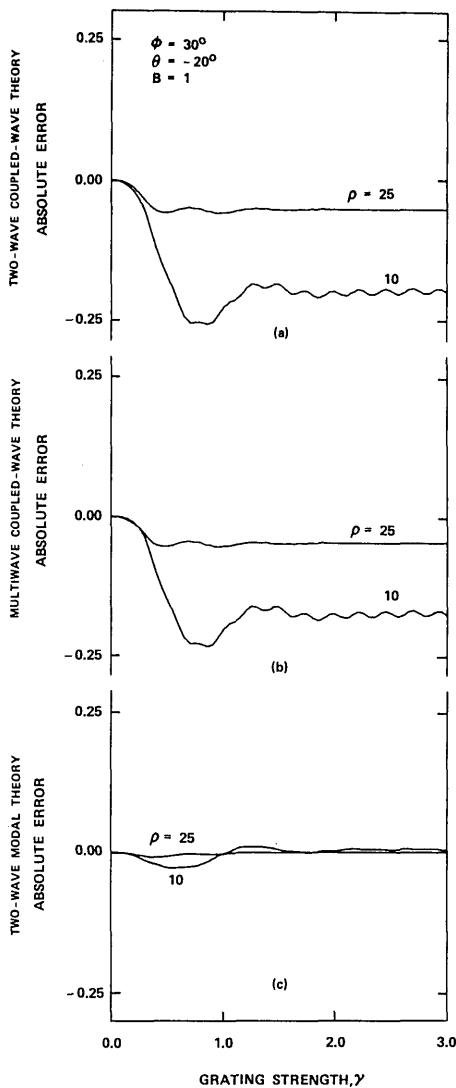


Fig. 9. Absolute error in the first-order diffraction efficiency as predicted by (a) two-wave coupled-wave theory, (b) multiwave coupled-wave theory, and (c) two-wave modal theory for the grating conditions shown in Fig. 4.

Incidence at the second-order Bragg condition is shown for a pure reflection grating in Fig. 6. It is important to note that solutions are not possible at even-order Bragg incidence ($B = 2, 4, 6, \dots$) unless second derivatives and boundary diffraction are included in the analysis. For even-order Bragg conditions the coefficient of the first derivative term in Eq. (6) vanishes for pure reflection gratings. Thus calculations such as those shown in Fig. 6 are not even possible with most approximate theories.

5. DISCUSSION AND CONCLUSIONS

The diffraction of a plane electromagnetic wave incident obliquely upon a planar grating bounded by two different media has been analyzed by using an exact and rigorous coupled-wave approach. The grating vector may have any arbitrary angle with respect to the boundaries. The solution has been formulated in terms of state variables in a simple matrix form that is easily implemented on a digital computer. Sample rigorous calculations have been presented for trans-

mission, reflection, and, for the first time, for general slanted gratings.

The present coupled-wave approach has been compared with the rigorous modal approach and with approximate modal and coupled-wave approaches. It was shown that inclusion of higher-order waves is more important than the inclusion of second derivatives and boundary diffraction for obtaining accurate predictions of diffraction efficiency when forward-diffracted waves are dominant (transmission-grating behavior). Conversely, it was shown that inclusion of second derivatives of the field amplitudes and boundary diffraction is more important for obtaining accurate predictions of diffraction efficiency when backward-diffracted waves are dominant (reflection-grating behavior).

The present method of analysis is useful in all applications in which planar gratings are utilized. However, it is especially valuable in applications in which slanted gratings and reflection gratings are used, such as in grating couplers and distributed-feedback lasers, since it is difficult to use the modal approach in these cases. This analysis may also be

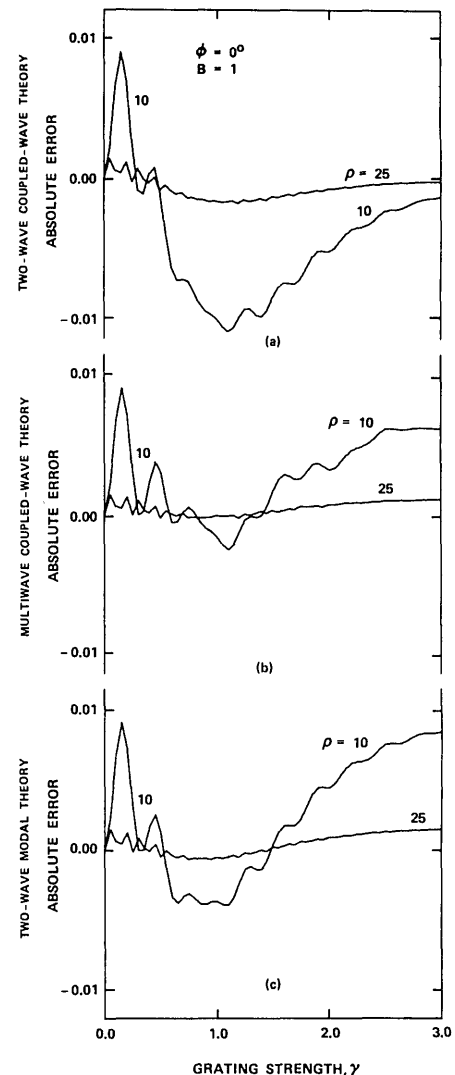


Fig. 10. Absolute error in the first-order diffraction efficiency as predicted by (a) two-wave coupled-wave theory, (b) multiwave coupled-wave theory, and (c) two-wave modal theory for the grating conditions shown in Fig. 5.

straightforwardly extended to absorption gratings and to mixed phase and absorption gratings, if desired.

APPENDIX: THE STATE EQUATION

The elements of the matrix $[b_{rs}]$ in Eq. (7), the state equation, are determined by Eq. (6), the governing set of differential equations. By defining the state variables as

$$S_{1,i} = S_i(u), \quad (A1)$$

$$S_{2,i} = \dot{S}_i(u), \quad (A2)$$

the constituent state equations may be written as

$$\dot{S}_{1,i} = S_{2,i}, \quad (A3)$$

$$\dot{S}_{2,i} = aS_{1,i+1} + b_i S_{1,i} + aS_{1,i-1} + c_i S_{2,i}, \quad (A4)$$

where $a = 8\epsilon_2/\Delta\epsilon$, $b_i = -a\rho i(i-B)$, and $c_i = a(\cos\theta' - i\mu \cos\phi)$. Writing Eqs. (A3) and (A4) as a single matrix state equation gives Eq. (7). In expanded form it is

$$\begin{bmatrix} \dot{S}_{1,2} \\ \dot{S}_{1,1} \\ \dot{S}_{1,0} \\ \dot{S}_{1,-1} \\ \dot{S}_{1,-2} \\ \vdots \\ \dot{S}_{2,2} \\ \dot{S}_{2,1} \\ \dot{S}_{2,0} \\ \dot{S}_{2,-1} \\ \dot{S}_{2,-2} \\ \vdots \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ \dots & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ b_2 a & 0 & 0 & 0 & 0 & c_2 & 0 & 0 & 0 & 0 \\ a & b_1 a & 0 & 0 & 0 & 0 & c_1 & 0 & 0 & 0 \\ \dots & 0 & a & b_0 a & 0 & 0 & 0 & c_0 & 0 & 0 \\ 0 & 0 & a & b_{-1} a & 0 & 0 & 0 & c_{-1} & 0 & 0 \\ 0 & 0 & 0 & a & b_{-2} & 0 & 0 & 0 & 0 & c_{-2} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix} \begin{bmatrix} S_{1,2} \\ S_{1,1} \\ S_{1,0} \\ S_{1,-1} \\ S_{1,-2} \\ \vdots \\ S_{2,2} \\ S_{2,1} \\ S_{2,0} \\ S_{2,-1} \\ S_{2,-2} \\ \vdots \end{bmatrix}. \quad (A5)$$

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